

PATHWAYS: CAMPUS CONNECTION MADE EASY

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Abstract

Pathways is a webapp designed to build meaningful connections between students and the resources available to them on campus by unifying navigational, academic, career, and social tools into a single platform. Focusing both on navigational and academic aspects of student life, this application bridges gaps that often hinder the student experience, offering features such as optimized route planning, centralized event discovery, calendar synchronization, career planning tools, and countless other tools that integrate seamlessly with existing academic systems. By consolidating these essential functions, Pathways enhances accessibility and organization across campus life. The system is powered by a Django REST Framework backend, with Celery and Redis enabling efficient asynchronous task processing, while a React-based frontend leverages Leaflet and the Leaflet Routing Machine for dynamic, interactive mapping. The application is fully containerized using Docker, served through Gunicorn and Nginx, and deployed on Render. Continuous integration and deployment are maintained via GitHub Actions, ensuring reliable automation in building and updating the platform. The GitHub for this project can be found [here](#), and our website is currently served at <https://www.pathways.sbs/>.

Problem Description

Information Fragmentation

Students in the University of South Carolina system and in universities all over the country currently rely on multiple different disconnected platforms in order to organize their student lives and find success in their academic and social pursuits. Commonly used sites like Blackboard and GarnetGate offer solutions to problems like course management and event planning, but are scattered across different websites and lack cohesion. This means that even though a student's day isn't neatly divided into categories, the apps and systems they rely on usually are. Class information lives in one place, events in another, navigation somewhere else, and other crucial tools like career planning tools and calendars lack synchronization. This makes planning and organization feel like a fragmented and chaotic process, since even though all of these tools exist, they rarely work together in a way that reflects how students actually move through their day.

Barriers to Learning

This fragmentation leads students to switch between these different apps multiple times a day and creates the need to juggle lots of information located in separate places, which can especially be detrimental to students not familiar with the demands and routines of college life. For freshmen, new transfer students, or even older students looking to explore new facets of campus for the first time, this lack of cohesion across university services can make a complex journey like college feel even more confusing, with students getting discouraged and lost in the shuffle easier. For example, the myriad of university events offered may be commonplace to a senior, but overwhelming to a freshman who doesn't understand what words like 'Baraza' or 'EQ' mean. Knowing that these are both buildings isn't intuitive, so students may not know where to route themselves if they'd like to attend an event at these places, leading them to miss out on a potential campus connection simply because of the lack of connection between GarnetGate and Google Maps.

Lost in the Shuffle

In this vast amount of information, important details get lost or forgotten by students; for course-related information this makes it more difficult for students to succeed academically, while confusion about events on campus or where important buildings are can contribute to a sense of alienation. For example, a student looking to get involved in campus research must go to the campus OUR database, keep separate note of the professors they're interested in, seek the professors' information out separately, and then draft each individual email by themselves. This inevitably leads to students

forgetting who they wanted to contact, having difficulty narrowing down what professors they'd like to contact based on their interests, and failing to achieve their main objective: getting involved with a professor whose skills align with their own. A single student experiencing this may seem trivial, but on a larger scale, this information leakage leads to thousands of student interests, deadlines, and paths to success falling through the cracks.

University-Wide Implications

While the consequences of not having a central campus-focused application mainly affects students, who face unnecessary stress and inefficiency, this problem also affects faculty and campus organizations that struggle to keep students informed and engaged. Faculty are often required to duplicate announcements across multiple platforms, increasing the likelihood that critical course information is overlooked, while student organizations face similar challenges in promoting events and maintaining consistent communication, resulting in missed opportunities and lower participation.

Taking this to an even higher level, we can see how this problem affects university leadership, since student success and retention improve when students feel supported and connected. When students miss deadlines, disengage from campus life, or feel overwhelmed by disorganization, it impacts both their academic performance and overall experience, which in turn affects key institutional metrics such as retention and satisfaction. The demands of a fast-growing university like USC elicit the need for a single place where students can come to manage and explore their student needs, one that offers a unified and intuitive platform that reduces fragmentation and strengthens connection across the entire campus community.

Pathways' Solution

General Overview

Our application focuses on building pathways between students and the resources they have available to them on our campus. Focusing both on navigational and academic aspects of student life, the intention of Pathways is to bridge the gap between students and the finer aspects of university life that can greatly improve the student experience. By bringing these tools into a single, intuitive platform, Pathways aims to streamline daily tasks and reduce the confusion caused by scattered systems, all while giving students all the tools they need to succeed both academically and mentally.

Navigational Tools

The Map feature provides an interactive campus map that supports directions and routing across

campus locations. It includes a hazard reporting system that allows users to flag issues in real time, with gamification through XP tracking where users earn 100 XP per report and 50 XP per upvote. Users can view XP badges and tooltips that reflect their activity and contributions over time. The feature integrates location services to enable accurate positioning and routing, while also supporting user interaction through map-based comments and reports that enhance overall situational awareness.

Campus Event Tools

The Events feature allows users to browse a dynamic event feed with built-in search and pagination for efficient navigation. Users can filter events by date range, time of day, and day of the week, as well as whether events are on or off campus. Event creators have additional controls to filter events by status, including pending, approved, or rejected submissions. Users can toggle between viewing all events and those they have created, save events to a personalized favorites list, and access detailed event pages. The feature supports category-based filtering and provides toast notifications to keep users informed of updates. Admins and event creators can monitor pending event counts and event-related analytics, which helps streamline event approval workflows.

Course and Academic Tools

The Courses feature provides a centralized space for managing academic information, including viewing courses and assignments and uploading structured course data through JSON. It includes a flexible post-it notes system that allows users to add, edit, remove, and pin notes for quick reference. Users can track assignment completion and engage with gamified elements such as weekly goals, daily streaks, on-time streaks, and rewards that encourage consistent progress. The feature also includes a grade dashboard that performs weighted calculations and projections, along with category-based breakdowns to give users a clearer understanding of their performance. Users can pin important assignments so they remain visible. Announcements are tracked so users can see which ones they have already viewed. Each course also allows grading to be customized based on the user's needs. Additionally, users can plan for semesters ahead by viewing their school's course catalog and adding and editing different plans they might have.

Calendar Tools

The Calendar feature offers a comprehensive scheduling system with multiple views, including day, week, and month layouts. Users can easily manage their schedules through event editing and can create, modify, or delete both single events and recurring event series. The feature supports custom

calendar organization with color coding and allows users to define event types such as academic, personal, campus, or custom categories. It also integrates with the map system to provide directions to event locations. Auto-generated event summaries help users quickly understand event details, while event completion tracking supports productivity. Users can import syllabus files in PDF, DOCX, or ICS formats, and the system includes reminders and filtering options to ensure important events are not missed.

Career Planning Tools

The Career feature is divided into Research and Industry sections to support different career pathways. The Research tab includes a ten-question research vibe quiz that matches users with professors based on their interests and responses. Users can save professors to a shortlist for later review and explore fellowship opportunities using advanced filtering tools. They can save fellowships, compare up to four options side by side, and organize opportunities into folders for better tracking. The Industry tab provides tools such as a career planner, mock interview preparation, and an application tracker to manage job applications. It also integrates with the Muse API to support job searching, giving users access to a broader range of opportunities within a structured interface.

Discussion Tools

The Discussion feature enables users to create and participate in discussion threads that can be either private or public depending on their needs. Users can invite others from the same university, while administrators have the ability to enable cross-university discussions. The feature supports message posting with markdown formatting, along with editing and deleting capabilities for both messages and threads. It includes real-time polling to check for new messages and provides toast notifications to keep users updated on activity. Moderation tools are available to maintain appropriate use, and users with the correct permissions can access invitation history to track participation and thread management.

Other Tools

The public homepage serves as a personalized dashboard that provides a quick overview of key academic and campus information. It displays a welcome message along with countdown timers for the next exam and next assignment, each showing days, hours, minutes, and seconds and linking directly to the assignments section. The dashboard includes an interactive campus map preview that supports dragging, zooming, scroll wheel control, and keyboard navigation. It also features a weekly workload heatmap that visually represents assignment density using a color scale from green to red.

Each day shows assignment counts, and users can click a day to expand a detailed list of assignments with direct links to the courses page. A legend explains the workload scale, and quick-access cards provide navigation to Courses, Calendar, Events, Career, and Settings.

The Settings Page provides centralized account management and user customization options. The Profile section allows users to view and update their name and email, with email changes requiring verification and showing pending status until confirmed. Users can also cancel pending email changes. The Security section includes password management with current and new password verification, as well as two-factor authentication options such as SMS-based codes and app-based authentication through QR codes or manual entry. Users can disable two-factor authentication after verification and view their current security status. The Accessibility section allows users to adjust font size and toggle the OpenDyslexic font, with preferences saved per account. The Preferences section lets users rank event categories, choose notification delivery methods, and toggle class reminder notifications. The Account section includes a role change request system for upgrading from User to Event Creator with a required justification, as well as an account deletion option with confirmation. A navigation sidebar provides quick access to all major settings sections, including Profile, Security, Accessibility, Preferences, and the Danger Zone.

Societal Considerations and Expectations

Ethical Considerations

User Privacy and Data Expectations

Pathways collects and processes several types of sensitive data including location data, student schedules, academic deadlines, event attendance, and campus movement. Students have a reasonable expectation of privacy regarding their academic data, campus movements, and personal identifiers.

In general, Pathways attempts to store minimal data on the user, and any additional data is opt-in to best preserve privacy. Additionally, users have access to a privacy policy that clearly lays out how their data is stored and used. More specifically, personal account data is kept private, and only data reasonably expected to be seen by other students (such as event postings, traffic reports, and other map comments) is shared with other users.

In terms of the map feature, the most important user data is location and route data. Pathways does not attempt to obtain a user's current location until they explicitly enable that option. As a further mitigation of any potential privacy issues, their current location and route are also stored client-side. Of course, if a user chooses to save routes to their account, they are stored, but only the

user is able to see their routes, and the saved routes can be deleted at any time. Additionally, if a user chooses to report or comment on the map, that data may be shown to other users, and this information is communicated to the user in the relevant areas.

For the events and calendar features, users expect that events that they publicly create are shown to others. However, events from the events page can also be added to a user's calendar. To preserve the user's privacy, event attendance and calendar events are only seen by the user that created the event.

Potential Discrimination and Mitigation Techniques

Pathways does not directly collect or process demographic information about users that would identify users as being part of a group that faces discrimination, so the software itself cannot explicitly discriminate against users. Additionally, any information that we collect from events the user attends is only used for the user's own calendar and events and is not used for anything else.

However, routing algorithms could unintentionally or implicitly discriminate by assuming certain defaults (e.g., walking speed or terrain tolerance) that exclude users with disabilities. To prevent this from happening, any page that uses maps or routing has an explicit accessibility mode that displays accessibility information on both maps and directions. Additionally, the site allows users to select their default method of transportation and adjust their default movement speeds.

Additionally, AI elements are used in this webapp, but only to tag certain events into categories. Using natural language processing to automatically generate category tags for campus events can unintentionally introduce bias because the algorithms used to analyze event descriptions learn from existing data, which may reflect historical bias. This may lead to overrepresenting certain types of events while underrepresenting others, like cultural gatherings or activism. To mitigate this, Pathways periodically audits tag distributions to identify bias and adjust model training data or tagging rules accordingly.

Finally, most of the content in this application is user-generated, so there may be some potential discrimination from other users and the content they generate. However, user-generated content is monitored under the Potential User Abuse section above, and any users engaging in discriminatory activities with Pathways are subject to bans or other disciplinary actions preventing them from interacting with users.

Potential User Abuse and Mitigation

Users could attempt to cause harm to the public at large by attempting to leak academic class

data. However, this is not really a problem because Blackboard and nearly every other class management software has the same issue, and if needed, we can prevent users from downloading any documents that have been obtained from Blackboard. Users can also attempt to cause harm to other users. There are multiple ways users communicate with other users and potentially abuse that communication to cause harm, namely posting events and making comments on the map.

Event posting could result in harm if an event's page contains inappropriate pictures or descriptions, hateful speech, links to untrustworthy sites, or false or defamatory information. Additional harm may come about if a user attempts to impersonate an organization and get students to come to fake events. To mitigate this, Pathways first requires event posters to have a verified student email and agree to clear community guidelines. Pathways also has an admin-based organization verification system for events. Furthermore, changes to events are tracked, so that users cannot cause harm and then attempt to hide it by editing their event. If a user posts content that breaks community guidelines, then they may be temporarily or permanently banned from creating events depending on the severity, and a moderator can delete their account or take disciplinary action.

Users can also abuse their ability to make map comments to cause harm, not only by the comments they post, but also in reporting false hazards to disrupt or mislead other users. In terms of comments that users produce, we implement the same content moderation and mitigation system as the one for event posting, including only allowing verified students to comment, adding keyword filtering/flagging, preventing flagged content from being seen by other users, and making users agree to the community guidelines before being allowed to participate. Additionally, if a user or users create false hazard reports or otherwise violate community guidelines, they may be temporarily or permanently banned from making map comments, reporting hazards, or even lose access to the map feature depending on the severity.

Legal Considerations

Intellectual Property Constraints

There are many intellectual property constraints placed by clients, which are universities seeking to provide Pathways for their students and faculty. In the case of intellectual property, there are four specific groups of individuals whose data must be protected.

Firstly, Pathways users who incorporate any borrowed imagery into their event postings must use copyright-free or Creative Commons licensed imagery while abiding by licensing conditions. However, if a user decides to create their own media for event creation purposes, the user owns the copyright to their imagery. Pathways follows university clients' copyright policies for media utilization. The

application's Terms of Service and User Agreement explicitly state this policy for users to observe. In addition, to safeguard users' original content that may be uploaded to the site, Pathways strongly enforces the Digital Millennium Copyright Act. If the administrators of the Pathways application receive notice of copyrighted materials being used for event postings, the event post is taken down. The Pathways website displays information regarding the importance of avoiding copyrighted imagery and how to notify administrators of copyrighted media on the website. Through compliance with copyright rules, university clients and the application avoid facing legal consequences.

Secondly, university clients provide their own branding for the customization of the application. Specifically, the University of South Carolina provides their logos, images, and wordmarks for personalization of the Pathways app. The university's brand is trademarked and there needs to be a license provided by the university to use it. All content using trademarks within the Pathways platform and branded external communications complies with the regulations set forth by the University of South Carolina's Trademark Licensing Office.

Additionally, professors have the copyright to their original teaching materials, which include lectures, assignments, and ideas. To be compliant with copyright laws, Pathways must have proper authorization from professors before using this content within the website. The University of South Carolina also has an intellectual property policy, which disallows the unauthorized use of any intellectual property held by faculty members or students. The policy protects IPs like academic content, new patents, and software. So, to adhere to this policy, Pathways only uses content that was previously authorized by its respective creator.

Lastly, integrating students' class schedules and enrollment information involves laws surrounding the Family Educational Rights and Privacy Act, otherwise known as FERPA. Although the website does not show or use sensitive data regarding grades, transcripts, or disciplinary records, it still uses course schedules and enrollment information which are considered protected education records. To adhere to FERPA, the website is limited to authorized users like students and designated university personnel. Pathways incorporates strict security procedures to protect users' data and prevent unauthorized access or disclosure.

Potential User Abuse and Legal Ramifications

The app, like many platforms that allow user interaction and content sharing, has the potential to be misused in ways that could break the law if the right safeguards are not implemented. For example, users might try to upload or share copyrighted materials, like images that they do not have permission to use. Others also post misleading or unauthorized information about campus events.

There is also a possibility that someone tries to exploit the platform to gather personal information from other users or distribute false reports. People can also try to use social engineering tactics to gain access to private data. Mitigations like app policies and/or university code of conduct help discourage this behavior, but these problems still remain significant issues that must be addressed.

To mitigate this, Pathways develops clear community guidelines that explain what types of content and behavior are prohibited, including any illegal activities, copyright infringements, and attempts to share harmful or deceptive information. These rules are supported by a detailed Terms of Service agreement that outlines users' legal responsibilities and the consequences of misuse. Before gaining full access to posting or sharing features, users are required to read and acknowledge these terms to ensure that they understand these rules when using the app.

In addition to policies, Pathways uses technical measures to help detect and prevent misuse. Moderation tools help identify inappropriate or unlawful content, and the team reviews flagged posts to determine if they should be removed or reported. All event submissions are reviewed to ensure content is appropriate prior to being shared publicly on the website. The platform also recognizes copyrighted material and prevents it from being uploaded without proper authorization. To protect against data theft or unauthorized access, the app uses secure login methods, encrypted data storage, and regular security reviews to identify and fix potential vulnerabilities.

Security Considerations

Sensitive Stored Information

The app collects some sensitive information, including emails, passwords, phone numbers, names, tokens, shared secrets, location data, IPs, and grades. To protect this information, the application only stores data for the minimum amount of time possible for the functionality of the app. In the case of location data, whether it is collected from GPS or from a manually specified start location, this is only for a single session of the route feature and is discarded as soon as the session ends. For tokens, they are naturally designed to be ephemeral because of their purpose.

Passwords are salted, peppered, and hashed so that if hashes are compromised, the attacker is not able to obtain a password. The encrypted login data is stored in the Django database, and environment variables are used to store the keys so that there is separation. Industry-standard asymmetric HTTPS is employed to protect user data while it is in transit so that packets cannot be sniffed by a tool like Wireshark.

Additional Protection Measures

Pathways prioritizes security measures, meaning that sensitive information such as user data and two-factor authentication secrets will be protected using encryption, with encryption keys stored securely in environment variables or local environment files excluded from version control. Security keys will be rotated periodically and the rotation events will be logged for accountability. Also, in order to confirm data recovery works as we intend it to, restoration tests will be conducted periodically. Finally, Celery handles high-load tasks asynchronously to avoid data corruption. Additional steps Pathways takes include emphasizing secure coding practices and using two-factor authentication for user and admin accounts.

Design and Architecture

Prototyping and Requirements Development

Before starting the coding and development phase of this project, the initial design of the webapp was mocked up using a high-fidelity prototype. The process began with a rough Figma design that focused on exploring overall structure, layout ideas, and how major features connect. This early version was intentionally simple and was used to test core navigation between areas such as the dashboard, events, courses, and map.

After this first iteration, the design was refined into a more detailed Figma prototype. This version was aligned more closely with the developing requirements and focused on improving user flow and usability across the platform. It clarified how users move between features and how interactions like filtering, event creation, and calendar management should feel during the user experience.

As we were completing the refined prototype, a full set of requirements was developed. These requirements included both core functionality and more advanced aspirational features. The system was then broken into smaller development milestones that directly inform how the architecture is implemented.

System Security and Core Backend

The system follows a monolithic architecture with separate frontend and backend services. The backend is built using Django 5.2 with Django REST Framework and exposes RESTful API endpoints that are consumed by the frontend. Communication between the frontend and backend occurs entirely through these APIs, allowing for a clean separation of concerns while maintaining a unified backend.

The backend uses a custom User model extending Django's AbstractUser. It runs on SQLite in devel-

opment and includes models such as User, Course, Assignment, CalendarEntry, Event, Discussion, HazardReport, and Route. Authentication is handled using JWT via `rest_framework_simplejwt`, with access tokens lasting 12 hours and refresh tokens lasting 1 day, with both being configurable. Token rotation with blacklisting is enabled for added security.

Password security is enforced using PBKDF2 hashing with a custom server-side pepper. The backend also tracks login activity including IP addresses. A middleware stack is used to handle security, sessions, CSRF protection, authentication, messaging, and frame options.

Frontend and Client-Side Architecture

The frontend is built with React 19.2 and React Router DOM 6.22 as a Single Page Application. It uses client-side routing and a component-based architecture to support modular feature development. State management is handled using React hooks and context, while local storage is used for user preferences and career match sessions.

A theme context system supports dark and light mode across the application. The frontend communicates with the backend through authenticated REST API calls using JWT tokens. Data is rendered dynamically without full page reloads, enabling an interactive user experience across the different tools that Pathways offers.

Infrastructure, Integrations, and Data Flow

The application is containerized using Docker Compose to ensure consistent development and deployment environments. Services include Redis for caching and task brokering, Django running with Gunicorn, Celery workers and schedulers, the React frontend, and Nginx as a reverse proxy.

Async task processing is handled through Celery with Redis. Scheduled tasks include pulling external event data, sending email digests, and refreshing user data at regular intervals. Caching is implemented using Redis via `django-redis` to improve performance for frequently accessed data.

External integrations include Mailgun for email services and Twilio for SMS-based authentication. We also use Google Gemini for AI-powered features such as event categorization and text generation. Event and job data are sourced from external APIs and feeds, while mapping is powered by OpenStreetMap, Leaflet, and GraphHopper routing.

Data flows from the frontend through authenticated API requests, where the backend validates tokens and processes logic, all while returning JSON responses. Background processing is handled asynchronously, while caching and polling support performance and near real-time updates for dy-

namic features such as discussions and notifications.

User Testing

Requirements and Proof of Concept Feedback

During early user testing, feedback on the requirements and initial design was generally positive. Testers noted that the feature set appeared comprehensive and addressed many common student pain points. The proof of concept demo reinforced this feedback, as it demonstrated that the core pages were functional and that the integration between features showed potential.

Users responded well to the idea of combining multiple campus tools into a single platform. The combination of maps, events, courses, and calendar tools was seen as valuable, and while the system was still in an early stage, testers indicated that the PoC was solid and that the application could become useful with further refinement and bug fixing.

Beta Milestone Feedback

At the beta milestone, more detailed usability feedback was collected. A key issue identified during this phase was that the application felt too ‘read-only’. Users reported that while they could view a large amount of information, they were not able to interact with or modify enough elements within the system. This reduced engagement and made some features feel incomplete.

In response to this feedback, improvements were made to increase interactivity across the platform. More editing capabilities were introduced, and additional user actions were enabled in areas such as courses, calendar management, the career page, and personal data. This helped shift the application from a more viewing-based experience to an interactive system.

Peer Feedback and Bug Resolution

Peer testing played a significant role in identifying usability issues and system bugs. Multiple issues were reported and resolved prior to the 1.0 release, both by the ColaCoders team and the team assigned to our project for the QA milestone. These included functional problems and missing features that affected overall usability.

Some notable issues included problems with the application tracker and deleting applications, incorrect placement of map indicators, and inconsistencies in event start and end times. Users also identified navigation issues, such as not having a clear way to return to the home page. Additional feedback highlighted functional failures in bike routing, inconsistencies in AI-generated helper in-

formation, general dark mode fixes and tweaks, and issues when multiple accounts were signed in simultaneously.

Other improvements focused on user experience. These included fixing broken button highlights, enabling login with email instead of only username, making sure that browser syncing worked, and clarifying the different user roles available within the system. Enhancements were also made to better clarify certain buttons and improve the course and assignment notes interface.

Backend Unit Testing

Backend unit testing was implemented to ensure the reliability and correctness of the system's core functionality. Testing was conducted using Django's TestCase framework along with pytest-django, and all tests were organized within the backend/testing directory. A total of 18 test files were developed to cover the major features of the application. Login functionality was verified for both username and email-based access, and session handling was tested to ensure consistency across requests. The custom user model was also validated to confirm that account creation and management followed the expected rules.

A large portion of testing focused on feature-level behavior across the application. Event-related tests ensured that filtering worked correctly and that only appropriate events appeared in user feeds. They also validated favoriting, review workflows, and administrative features such as pending event counts and creator analytics. Course and calendar functionality were tested to confirm that academic data could be stored, retrieved, and displayed without issues. Discussion features were tested at both the model and API level to ensure that threads and messages behaved correctly. Map functionality was also tested to verify that hazard reports and routing data were handled properly.

Additional tests focused on more advanced systems such as career tools and permissions. The career roadmap feature was tested to ensure that all role and year combinations were supported and that milestone data was structured correctly. Interview scoring and career recommendation nudges were also validated for consistency. API permission testing ensured that role-based access control was enforced across endpoints so that users could only access appropriate features.

Frontend Unit and Behavioral Testing

Frontend unit testing was implemented using Jest and React Testing Library to ensure that individual components and pages behaved correctly under different conditions. All tests were located in the frontend/src/tests directory, with 12 test files covering major areas of the application. The Home Page received testing to verify personalized welcome messages, countdown timers, and workload

heatmap interactions. Tests also confirmed that feature cards linked to the correct pages and that loading states and API failures were handled properly. Other pages such as Events, Courses, Calendar, Career, Discussions, Map, and Settings were tested to ensure that core functionality rendered correctly and responded to user input as expected.

The testing approach relied on mocking external dependencies to isolate component behavior. API calls were mocked to simulate both successful and failed responses. Libraries were also mocked to prevent external factors from affecting test results. Local storage and session storage were simulated to verify persistence-related features. These tests ensured that the frontend behaved consistently and that components made the correct API calls while responding properly to loading and error states.

In addition to unit testing, end-to-end behavioral testing was conducted using Playwright. These tests simulated real user interactions across the full application. Authentication flows were tested thoroughly, including login, registration, email verification, password reset, and two-factor authentication. Core features such as the homepage, calendar, courses, events, career tools, discussions, and map functionality were tested in a real browser environment. External integrations such as Mailgun were also validated by confirming that emails were sent and received correctly.

Potential Significance and Application

The importance that the Pathways app brings to USC and campuses like it is significant for multiple parties. For students, the immediate benefit is a reduced cognitive load and eliminating the need to switch between platforms, hopefully leading to higher academic success due to better organization and better campus mobility, especially for new students and those with accessibility needs. Additionally, Pathways provides students with a more efficient way to plan their day, synchronizing the academic and social elements of their life while prioritizing connection, improving the mental health of our student body. This consolidation of tools also helps students build more consistent routines, which can lead to better time management habits and reduced stress during high-pressure academic periods.

Pathways also gives faculty an improved way to communicate with students, since reminders, assignments, and announcements are more centralized and visible in our webapp. This will lead to higher student engagement and attendance since course information is easier to access, and reduces the need for professors to repeat assignment information or clarify missed assignment related issues. It also creates a more transparent academic environment where expectations are clearly communicated, which can improve the overall quality of interaction between students and instructors.

For campus organizations, Pathways provides a way to increase event visibility and turnout, giving smaller organizations the same platform as larger ones and allowing events to be pushed directly into student's calendars, increasing outreach. This helps foster a more inclusive culture where a wider range of student groups can gain recognition and participation, rather than being overshadowed by larger, more established organizations. It also encourages students to explore new interests and communities that they might not have otherwise discovered.

On a larger scale, university administration could benefit from an increase in student success, retention, and enrollment, due to students who now go to school in an environment where they feel more supported. The platform also provides opportunities for data-informed decision making, as aggregated usage patterns can highlight which services students rely on most and where additional support may be needed. Over time, this can help universities refine their student experience strategies and better allocate resources to areas that have the greatest impact.

What's Next

In the future, Pathways has broader significance as a foundation for USC's central student experience platform, which could either supplement or replace more fragmented services currently offered. Future versions could incorporate shuttle tracking, dining information, library capacity, and safety alerts, creating a central hub for all aspects of campus life. These additions would continue to reduce the need for students to rely on multiple disconnected systems and instead provide a unified interface for navigating daily campus activities.

Additionally, the dynamic design of the platform allows it to be deployed at other universities facing similar challenges with fragmented campus management applications. Many institutions struggle with separate systems for academics, events, transportation, and student engagement, and Pathways provides a flexible framework that can be adapted to different campus infrastructures. By adding campus-specific data dealing with core features such as mapping, research professors, and event management, Pathways can scale beyond USC Columbia and USC Beaufort and support a wider network of campuses.

Expanding to other campuses would also create opportunities for cross-university collaboration and shared improvements to the platform. Features such as event systems, discussion tools, and career resources could be extended to support broader student communities while still maintaining campus-specific customization. This scalability ensures that Pathways is not only a solution for USC, but also a model for improving the student experience across higher education as a whole.

Development Timeline

The ColaCoders team met every Tuesday and Thursday at 4:15PM during the fall semester, and every Wednesday at 9AM during the spring semester. Meetings typically consisted of a discussion of upcoming milestones and the deliverables due for that week, and were typically a mix of brainstorming our next steps and deciding how to delegate our tasks.

Milestone	Date	Tasks
<i>Form Teams</i>	Aug. 20th - Sep. 7th	Deciding what we want our project to be and what format we want to develop it in.
<i>Personas</i>	Sep. 7th - Sep. 14th	Developing who we want to build the app for and outlining their needs.
<i>Design</i>	Sep. 14th - Sep. 28th	Mocking up a high-fidelity prototype in Figma of the major screens of our app.
<i>Requirements</i>	Sep. 28th - Oct. 12th	Developing a comprehensive list of the features our app will include and implement.
<i>Architecture</i>	Oct. 12th - Oct. 19th	Detailing the framework and technical structure of our app (what languages, packages, API endpoints, etc.)
<i>Source Control</i>	Oct. 19th - Oct. 26th	Developing the initial codebase of our app and ensuring that all group members can interact with the GitHub.
<i>Ethical, Legal, and Security</i>	Oct. 26th - Nov. 2nd	Formulate ethical, legal, and security issues our webapp might encounter and develop solutions to ensure the management of these issues.

Milestone	Date	Tasks
<i>Proof of Concept</i>	Nov. 2nd - Dec. 5th	Develop an initial implementation of our webapp, using all planned APIs and implementing 10% of final app functionality. Film a short demo video showcasing the features of the PoC.
<i>Peer Reviews</i>	Nov. 28th - Dec. 8th	Give feedback on the contributions and efforts of our group members and discuss their impact on the project.
<i>Testing</i>	Jan. 12th - Feb. 2nd	Implement and run unit and behavioral tests to catch bugs and flaws within the system.
<i>Beta Release</i>	Feb. 2nd - Mar. 2nd	Develop a minimum viable product and have all major features implemented within the webapp.
<i>RC1 Release</i>	Mar. 2nd - Apr. 6th	Implement all features of the apps without any bugs and make sure the webapp is a cohesive, final product.
<i>Quality Assurance</i>	Apr. 6th - Apr. 20th	Complete thorough testing of another team's webapp and notify them of bugs present in their app.
<i>Marketing Webpage</i>	Apr. 6th - Apr. 24th	Create a webpage that shows off the features of our webapp and showcases its benefits to potential users.

Milestone	Date	Tasks
<i>1.0 Release</i>	Apr. 6th - Apr. 28th	Implement all features of the apps and fix all bugs, developing the app into a fully finished, professional project.
<i>Final Demo</i>	Apr. 6th - Apr. 28th	Create a video fully showcasing the abilities of the app and describe the functionality, use-cases, and major features of the app.

Appendix

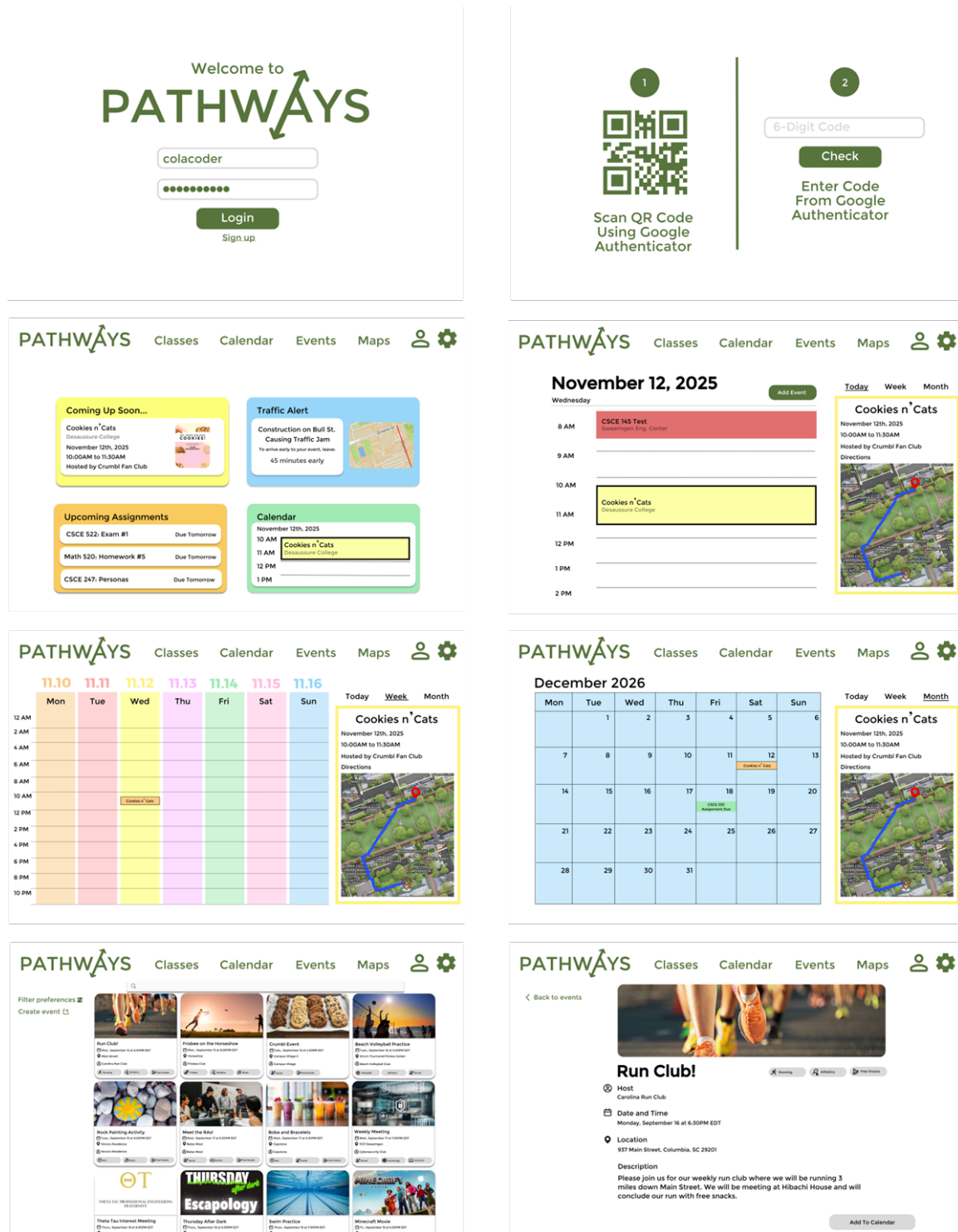


Fig. 1. Preliminary Pathways design screens (September 2025, Design Milestone)

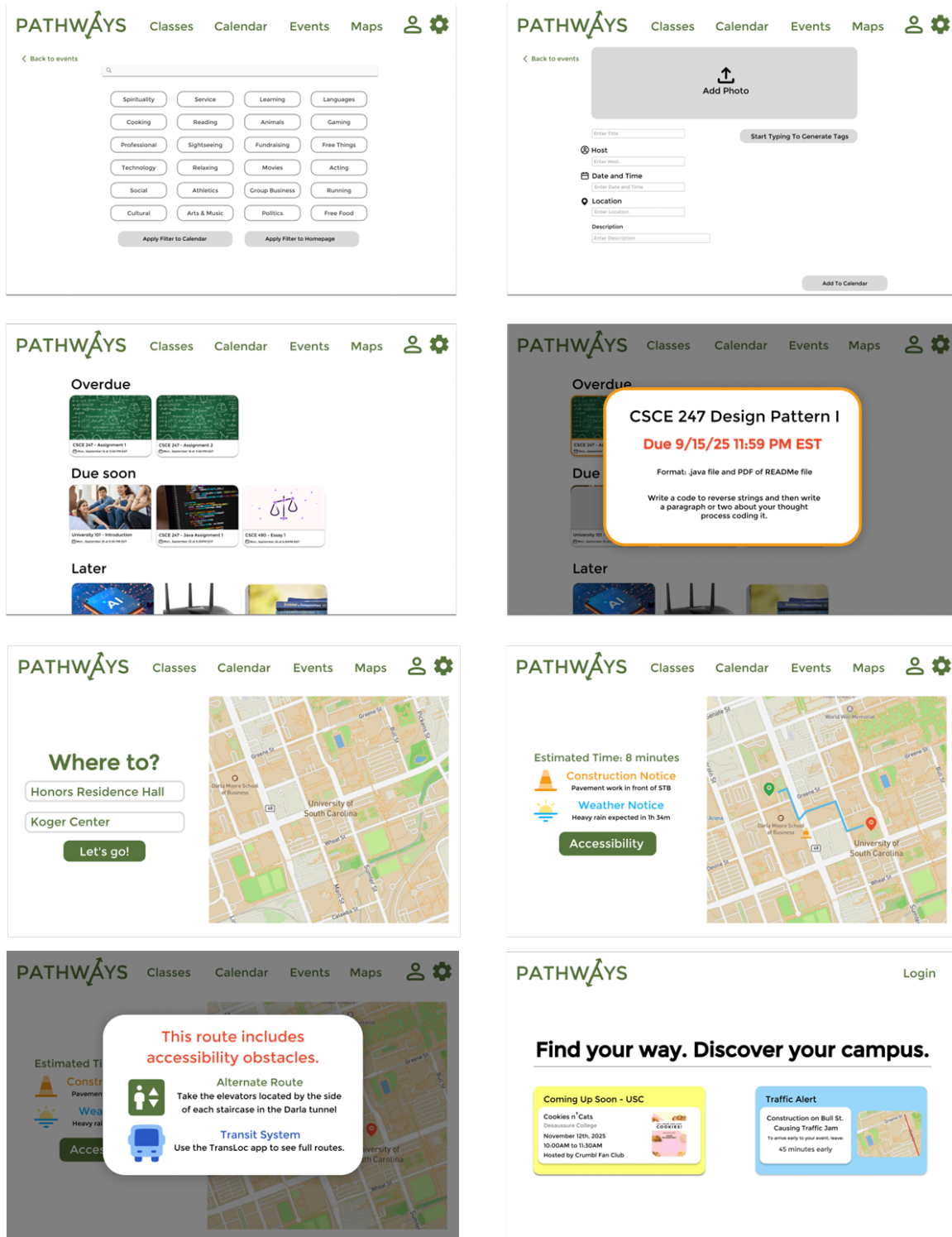


Fig. 1. (cont.) Preliminary Pathways design screens (September 2025, Design Milestone)

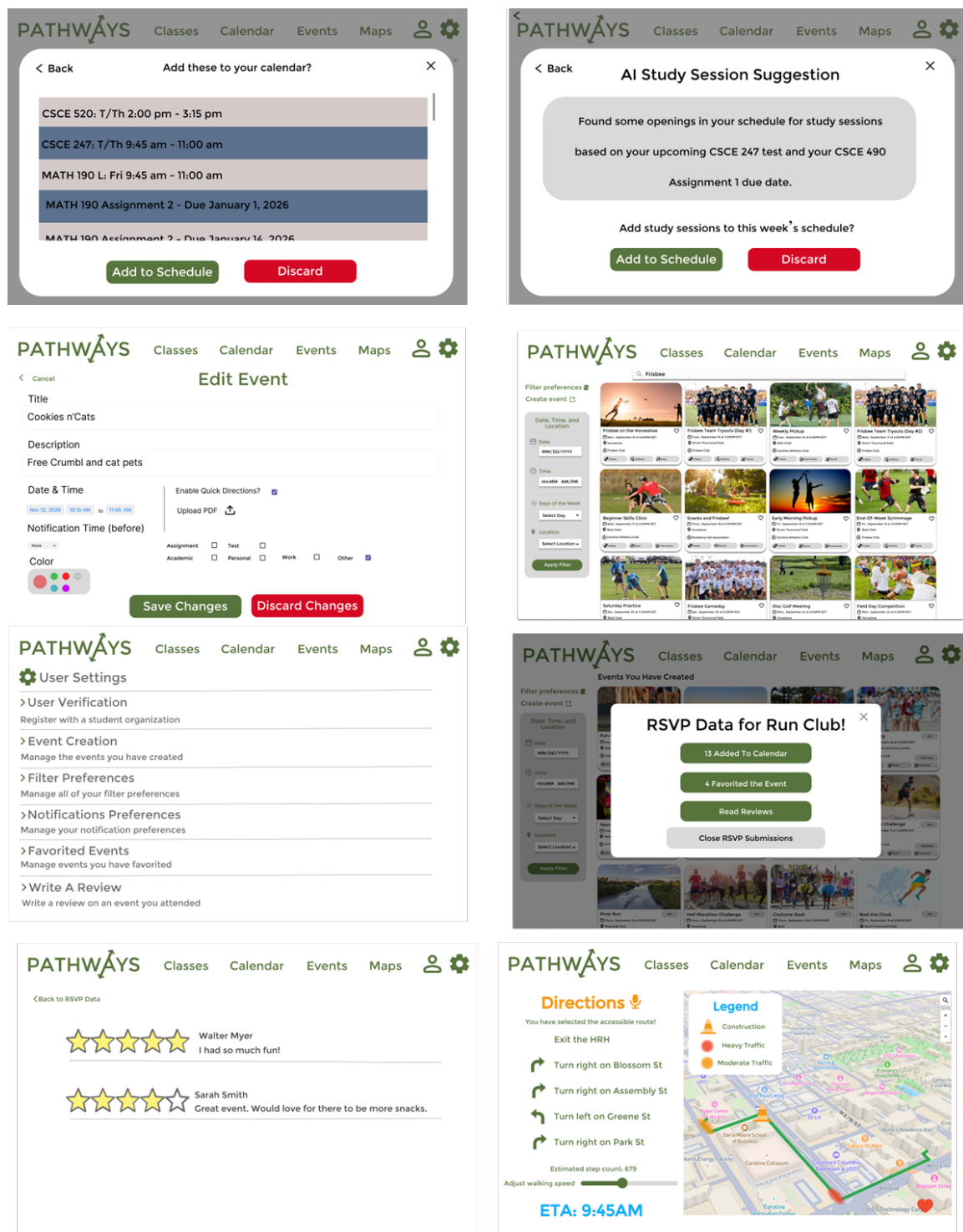


Fig. 2. Extended Pathways design screens (October 2025, Requirements Milestone)

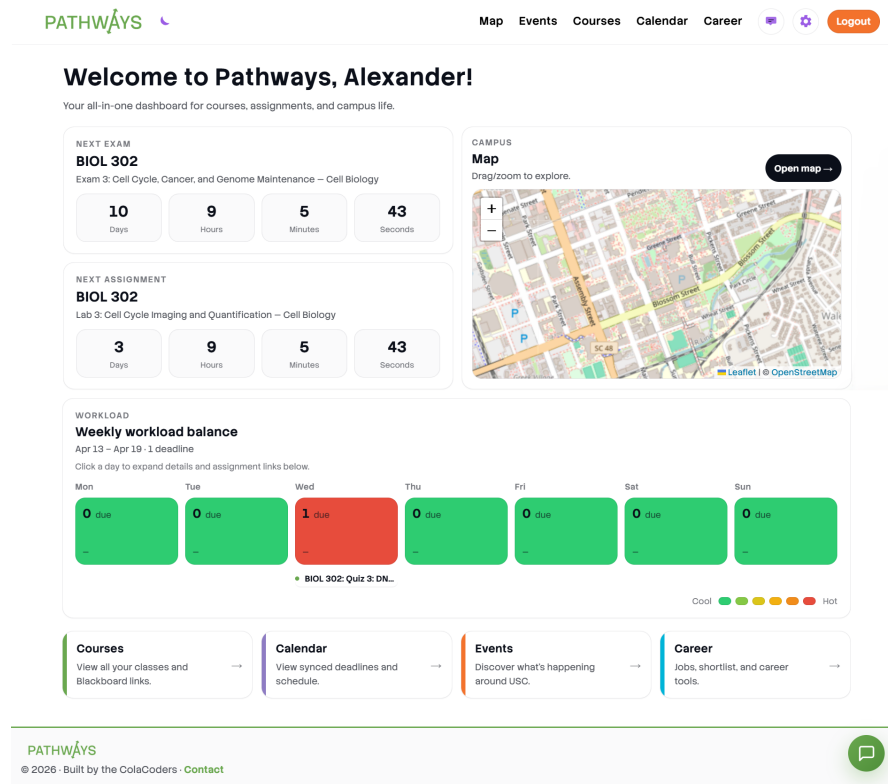


Fig. 3. Private, logged-in homepage

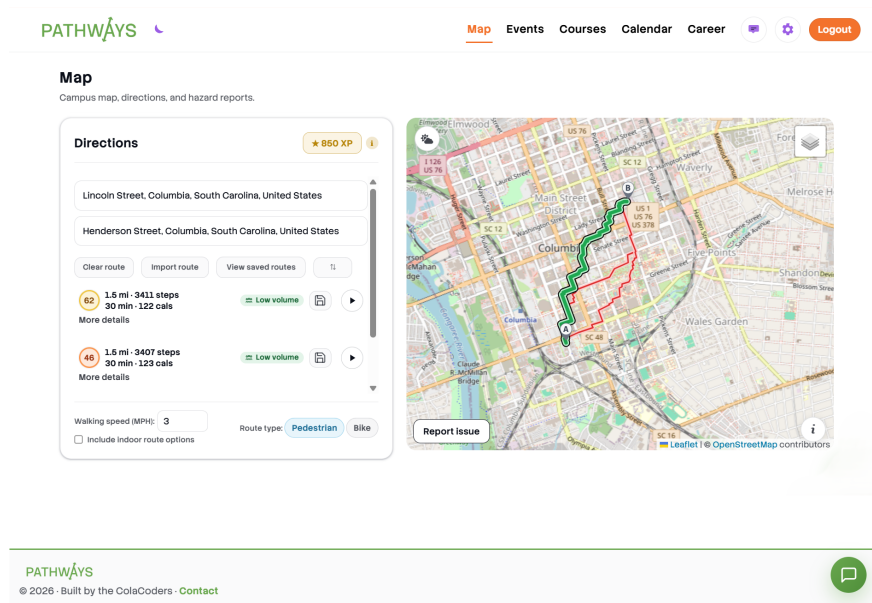


Fig. 4. Maps page with sample route loaded

PATHWAYS

[Map](#) [Events](#) [Courses](#) [Calendar](#) [Career](#) [🗨](#) [⚙](#) [Logout](#)

Events

Discover upcoming events, filter by date and time, and add them to your calendar.

Search for events!



Sunday Service
Sunday, April 19th at 10:00 AM
Location: Russell House Rm 315
Organization: Restored Campus Ministry
Categories: Spirituality, Social



Puppy Playtime
Sunday, April 19th at 2:00 PM
Location: East Quad Lobby
Organization: East Quad
Categories: Social



Sunday Supper
Sunday, April 19th at 6:00 PM
Location: St. Thomas More Student Center
Organization: The Catholic Campus Ministry
Categories: Social, Spirituality, Free Food



Worship and Free Dinner
Sunday, April 19th at 6:00 PM
Location: MSN Building
Organization: Methodist Student Network
Categories: Spirituality, Free Food, Social

« Previous 1 Next »

Show Events Created By Me

Filter Preferences

Favorited Events

Notifications 5

Write A Review

Create Event

Analytics

Event Filters

Date Range

Select A Date Range

Time

All Times

Days

All Days of the Week

Location

All

Clear Applied Filters

PATHWAYS

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Fig. 5. Campus events feed view for an event creator

25

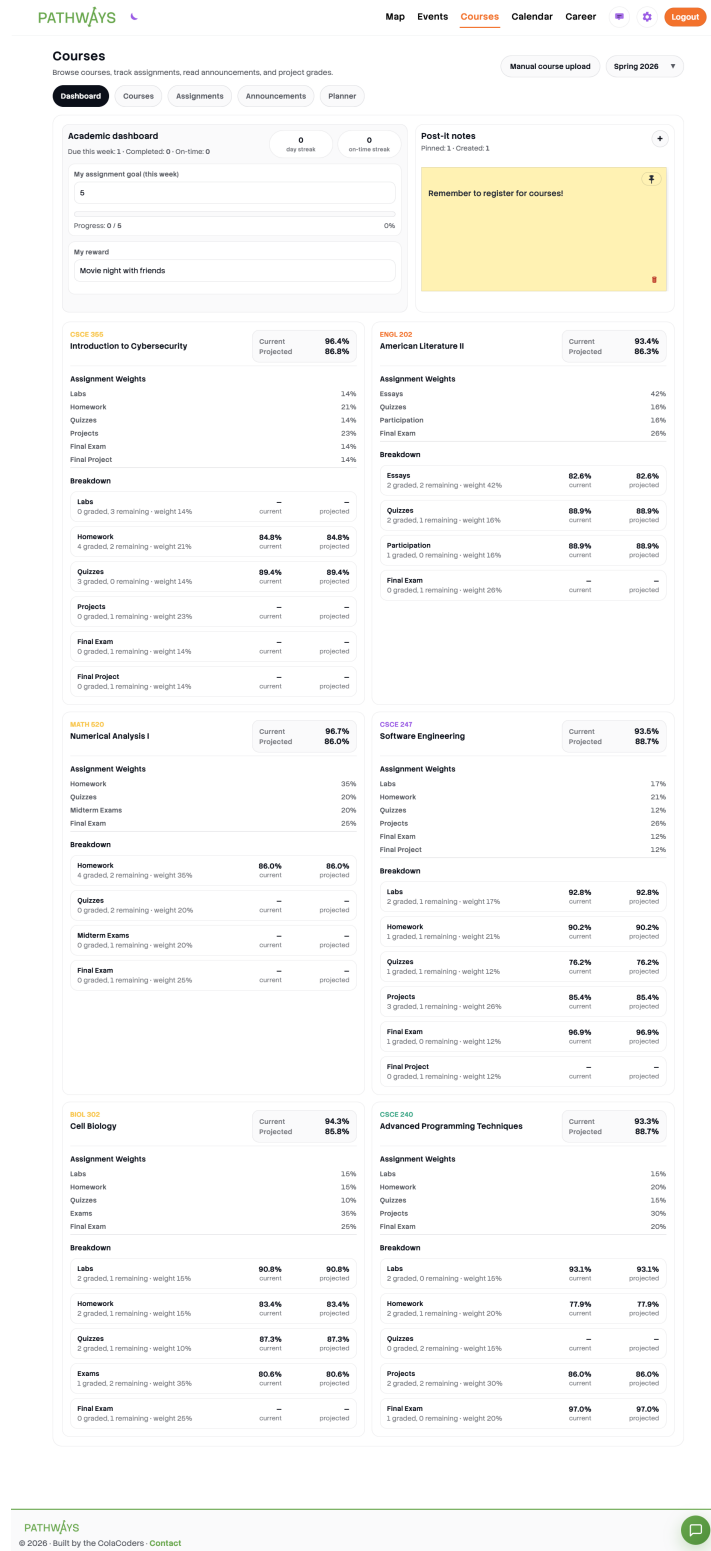


Fig. 6. Courses dashboard and academic overview

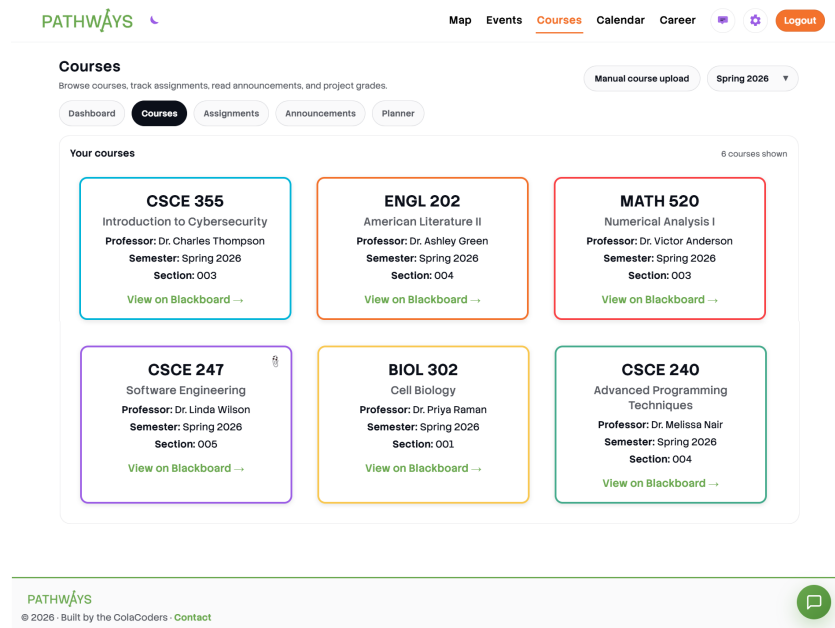


Fig. 7. General courses overview

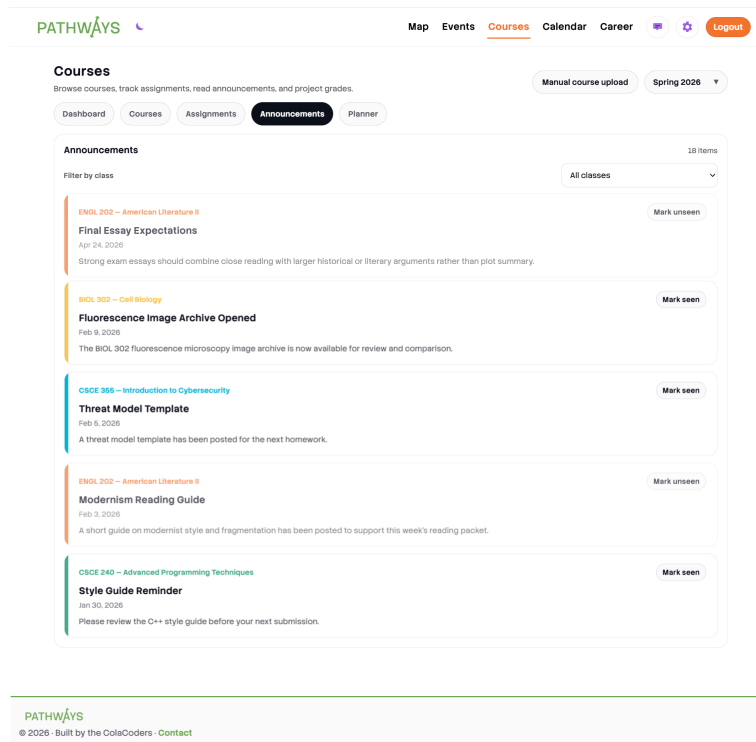


Fig. 8. Course announcements view

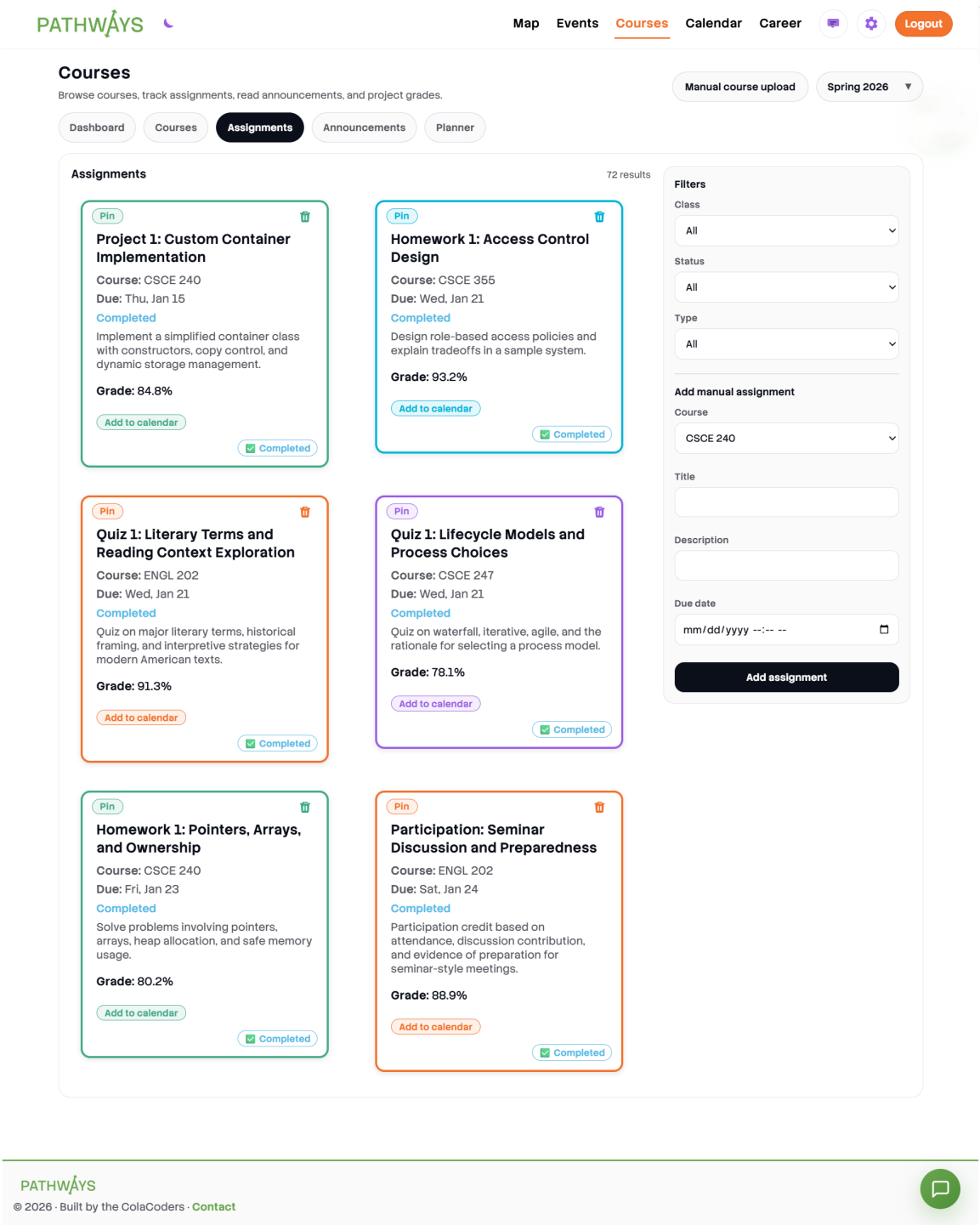


Fig. 9. Course assignments view

The screenshot displays the 'PATHWAYS' web application interface. At the top, a navigation bar includes 'Map', 'Events', 'Courses' (highlighted), 'Calendar', 'Career', and a 'Logout' button. Below the navigation bar, the 'Courses' section is active, showing a sub-header 'Browse courses, track assignments, read announcements, and project grades.' and a 'Manual course upload' button. The 'Planner' tab is selected, showing a list of courses categorized by semester: BIOL (3), CHEM (2), and PHYS (2). Each course entry includes the course ID, title, a description, and a 'Favorite' button. The 'BIOL 101' course is marked as '★ Favorited'. To the right of the course list, there is a 'Custom plans' section with a 'New plan name (e.g. Fall 2026)' input field and a 'Create plan' button. Below this, the 'My plans' section shows 'No plans yet.' and the 'My favorites' section lists 'BIOL 101 - Principles of Biology I'. The footer of the application shows the 'PATHWAYS' logo, copyright information '© 2026 - Built by the ColaCoders - Contact', and a green chat bubble icon.

PATHWAYS

Map Events **Courses** Calendar Career Logout

Courses
Browse courses, track assignments, read announcements, and project grades.

Manual course upload Spring 2026

Dashboard Courses Assignments Announcements **Planner**

Planner Favorites: 1 • Planned: 0

▼ BIOL (3)

BIOL 101
Principles of Biology I
Create a plan to start adding courses. ★ Favorited

BIOL 102
Principles of Biology II
Create a plan to start adding courses. ★ Favorite

BIOL 302
Cell Biology
Create a plan to start adding courses. ★ Favorite

▼ CHEM (2)

CHEM 111
General Chemistry I
Create a plan to start adding courses. ★ Favorite

CHEM 112
General Chemistry II
Create a plan to start adding courses. ★ Favorite

▼ PHYS (2)

PHYS 201
General Physics I
Create a plan to start adding courses. ★ Favorite

PHYS 202
General Physics II
Create a plan to start adding courses. ★ Favorite

Custom plans
New plan name (e.g. Fall 2026)
Create plan

My plans
No plans yet.

My favorites
BIOL 101 - Principles of Biology I

PATHWAYS
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Fig. 10. Course semester planner

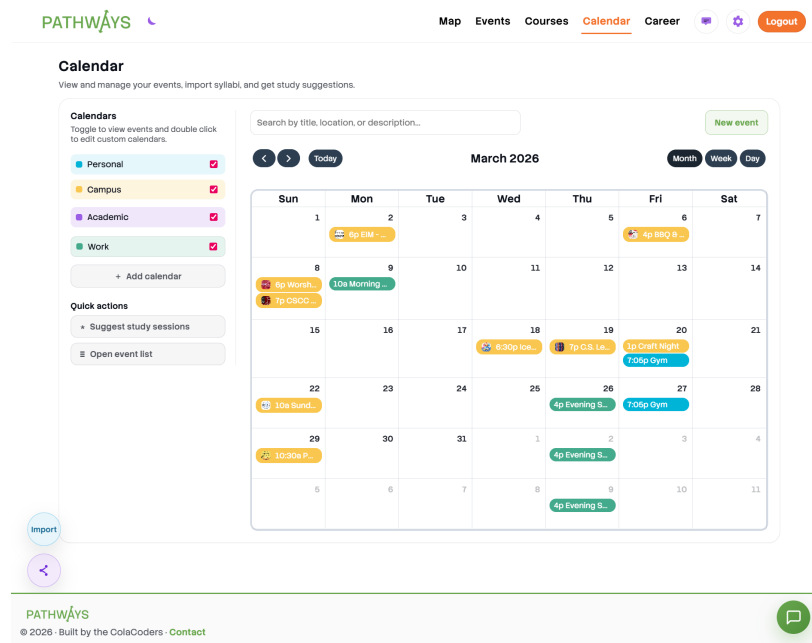


Fig. 11. Monthly calendar view with personal, academic, campus, and work events

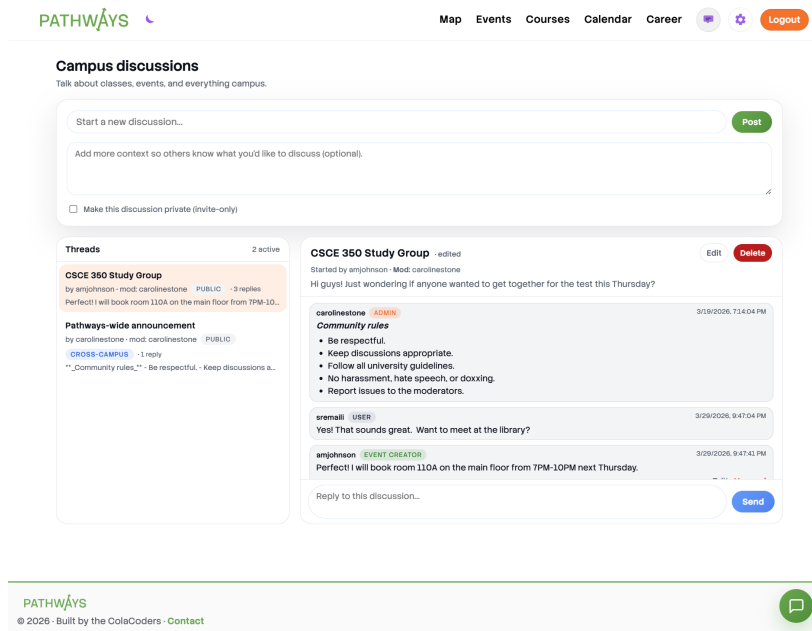


Fig. 12. Intra-campus discussions page

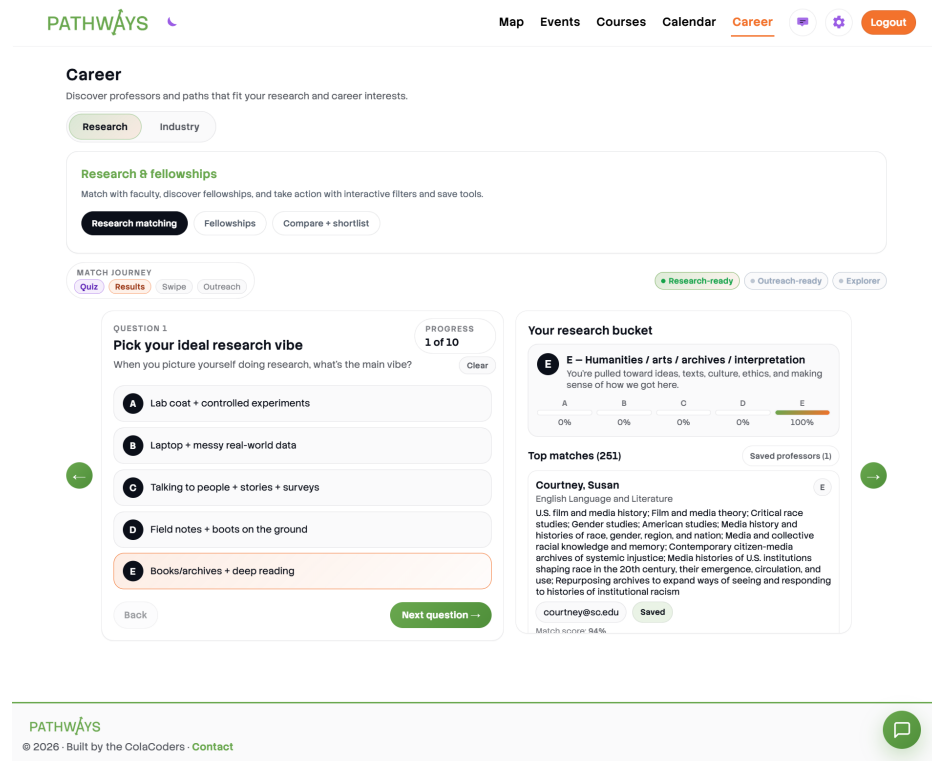


Fig. 13. Career page research matching quiz

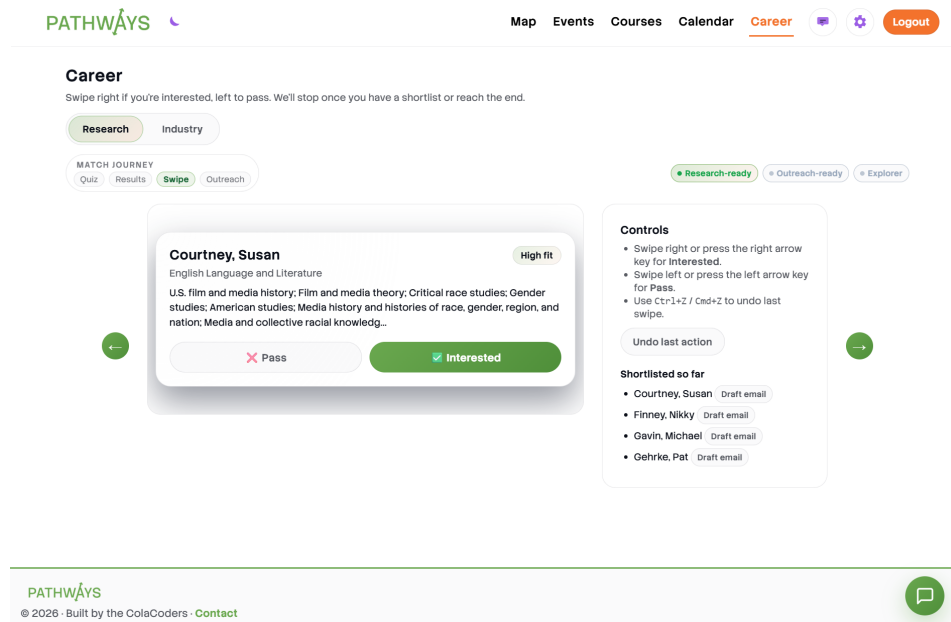


Fig. 14. Career page research matching swipe feature

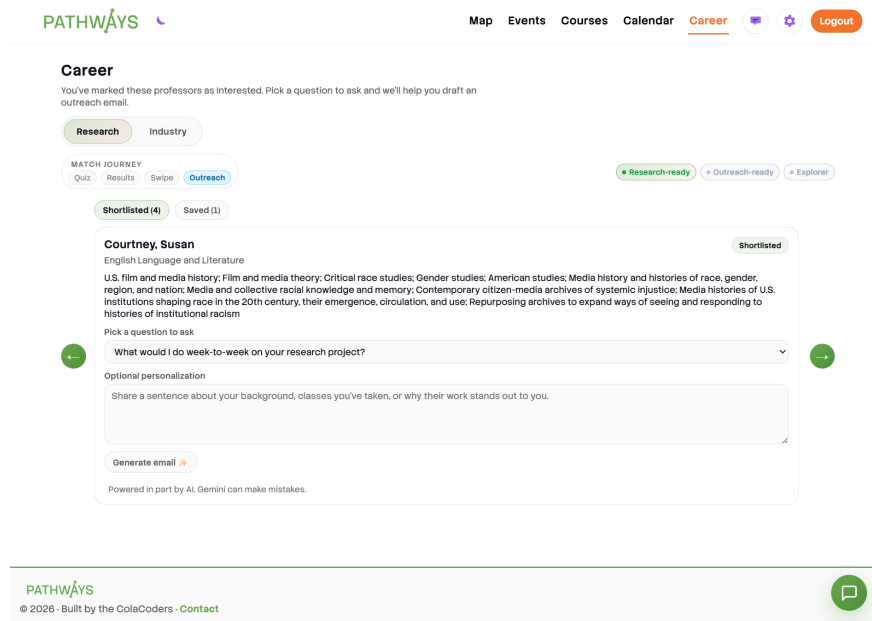


Fig. 15. Career page research matching email outreach generator

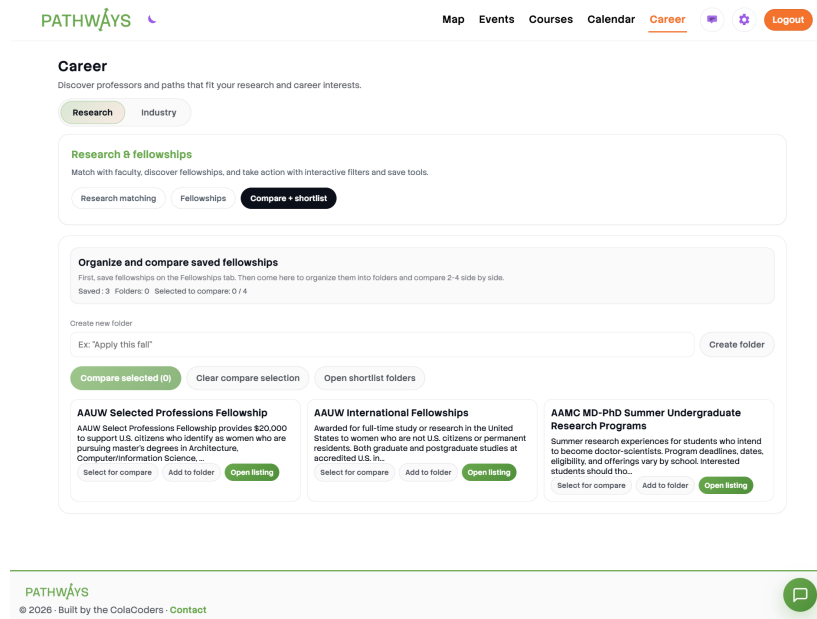


Fig. 16. Career page fellowship search compare and save

[Map](#)
[Events](#)
[Courses](#)
[Calendar](#)
[Career](#)
[User icon](#)
[Settings icon](#)
[Logout](#)

Career

Discover professors and paths that fit your research and career interests.

[Research](#)
[Industry](#)

Research & fellowships

Match with faculty, discover fellowships, and take action with interactive filters and save tools.

[Research matching](#)
[Fellowships](#)
[Compare + shortlist](#)

Saved
Surprise me

AAAS Mass Media Science & Engineering Fellowship

Ten-week summer program places science, engineering, and mathematics students at media organizations nationwide. Fellows use their academic training as they research, write, and report today's headlines, sharpening their...

Min GPA 3.0 No nomination Updated June 25, 2025

[View details](#)
[Open listing](#)

AAAS MD-PhD Summer Undergraduate Research Programs

Summer research experiences for students who intend to become doctor-scientists. Program deadlines, dates, eligibility, and offerings vary by school. Interested students should thoroughly investigate the websites of the...

Min GPA 3.2 No nomination Updated May 29, 2025

[View details](#)
[Open listing](#)

AATG Scholarships for German Studies

The American Association of Teachers of German (AATG) supports several scholarships for college students of German and members of Delta Phi Alpha. Be sure to check the competition website for the current cycle's deadline...

Min GPA 3.0 Needs nomination Updated June 20, 2025

[View details](#)
[Open listing](#)

AAUW American Fellowships- Dissertation Fellowship

Fellowship for women in all fields of study, to be used for the final year of dissertation writing. Dissertation Fellowships offset a scholar's living expenses while she completes her dissertation. The fellowship must be...

Min GPA 3.5 No nomination Updated June 19, 2025

[View details](#)
[Open listing](#)

AAUW International Fellowships

Awarded for full-time study or research in the United States to women who are not U.S. citizens or permanent residents. Both graduate and postgraduate studies at accredited U.S. institutions are supported. Recipients are...

Min GPA 3.5 No nomination Updated June 19, 2025

[View details](#)
[Open listing](#)

AAUW Selected Professions Fellowship

AAUW Select Professions Fellowship provides \$20,000 to support U.S. citizens who identify as women who are pursuing master's degrees in Architecture, Computer/information Science, Mathematics, and Statistics. Applicants...

Min GPA 3.3 No nomination Updated May 30, 2025

[View details](#)
[Open listing](#)

ABF/LAS/NSF Doctoral Fellowship Program in Law & Inequality

Supports original empirical and interdisciplinary research on the study of law and inequality. Applicants should be Ph.D. candidates who have completed all doctoral requirements except the dissertation. Doctoral research...

Min GPA 3.3 No nomination Updated June 10, 2025

[View details](#)
[Open listing](#)

ACFE Ritchie-Jennings Memorial Scholarship Program

Provides a one-time scholarship to students with a major or minor in accounting, business administration, finance or criminal justice and who wish to pursue a career in an anti-fraud profession. To be eligible to apply...

Min GPA 3.0 No nomination Updated May 30, 2025

[View details](#)
[Open listing](#)

ACLS Leading Edge Fellowships

Postdoctoral fellowship placing recent PhDs in the humanities and humanistic social sciences in two-year positions with nonprofit organizations. The fellowship aims to promote diverse career outcomes for humanities PhDs...

No nomination Updated June 19, 2025

[View details](#)
[Open listing](#)

Actuary of Tomorrow Stuart A. Robertson Memorial Scholarship

This scholarship recognizes and encourages the academic achievements of undergraduate students pursuing a career in actuarial science. Applicants must have successfully completed at least two actuarial exams and plan to...

Min GPA 3.0 No nomination Updated May 30, 2025

[View details](#)
[Open listing](#)

Advanced Fellowship in Health Services Research

Two-year opportunity for recently graduated MD/PhDs and other health professionals to conduct research and participate in clinical care in the Minneapolis VA Health Care System. The Center for Care Delivery and Outcomes...

No nomination Updated June 2, 2025

[View details](#)
[Open listing](#)

AH&LEF Arthur J. Packard Memorial Scholarship

This scholarship is awarded to the most outstanding student of lodging management and to two runners-up through an annual competition among Foundation-affiliated four-year programs. Deadline is typically in mid-March, bu...

Min GPA 3.5 No nomination Updated June 2, 2025

[View details](#)
[Open listing](#)

AICPA Foundation Scholarship for Future CPAs

Provides financial assistance to outstanding students to encourage their pursuit of accounting as a major and their ultimate entry into the profession. This award was established to support undergraduate or graduate stud...

Min GPA 3.0 No nomination Updated June 2, 2025

[View details](#)
[Open listing](#)

AIS Entrepreneurial Innovation Fellowship

Supports PhD students studying the intersection of innovation, entrepreneurship, and information systems. Funding supports study and research on topics that build on entrepreneurial-focused IS research streams (e.g. exam...

Min GPA 3.3 No nomination Updated June 2, 2025

[View details](#)
[Open listing](#)

Albright Institute Fellowships

Offers multiple fellowships to students and scholars in Near Eastern studies from prehistory through the early Islamic period, including the fields of archaeology, anthropology, art history, Bible, epigraphy, historical...

Min GPA 3.3 No nomination Updated June 2, 2025

[View details](#)
[Open listing](#)

Albritton Journalism Institute Reporting Fellowship

The AJI Reporting Fellowship provides 18 - 24 months of training and mentorship for aspiring journalists. AJI is committed to unbiased political journalism and restoring the public's trust in news media. Candidates of all...

No nomination Updated June 2, 2025

[View details](#)
[Open listing](#)

Allen Lee Hughes Fellowship

Professional training pipeline program to cultivate the next generation of theatre professionals. The program, named after the talented Associate Artist/Lighting Designer, Allen Lee Hughes, offers the highest standard of...

Min GPA 3.0 No nomination Updated June 2, 2025

[View details](#)
[Open listing](#)

Alpha Lambda Delta - Graduate Fellowship

Funding for qualified members of ALD to obtain graduate or professional degrees. Alpha Lambda Delta (ALD) is an invitation-only honors society for first-year academic success. Applicants must be inducted members who have...

Min GPA 3.5 No nomination Updated June 3, 2025

[View details](#)
[Open listing](#)

[Show more fellowships](#)

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Fig. 17. Career page fellowship search listings

33

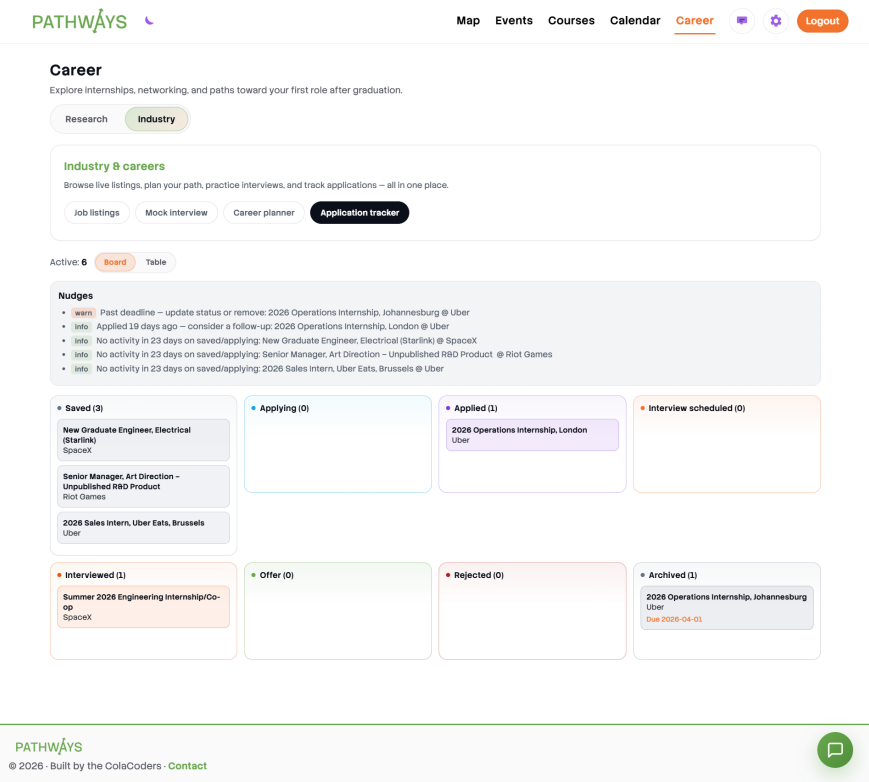


Fig. 18. Career page industry application tracker

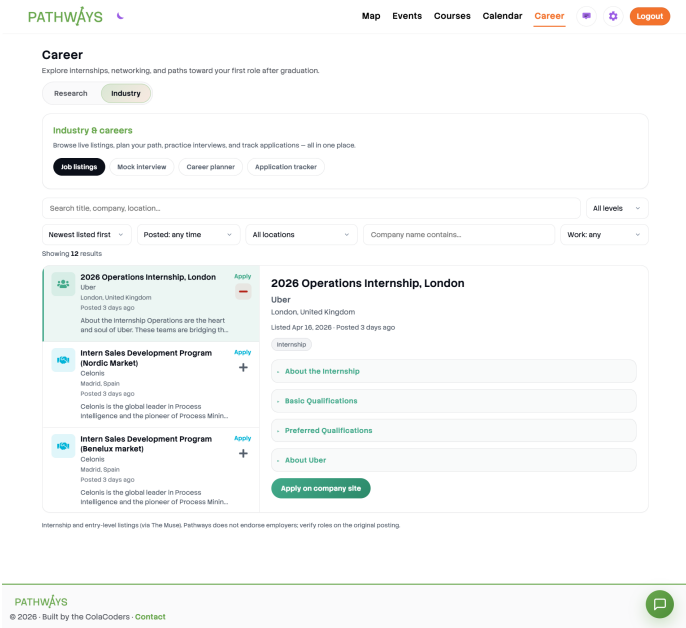


Fig. 19. Career page industry job listings



[Map](#) [Events](#) [Courses](#) [Calendar](#) [Career](#)   [Logout](#)

Career

Explore Internships, networking, and paths toward your first role after graduation.

[Research](#) [Industry](#)

Industry 8 careers

Browse live listings, plan your path, practice interviews, and track applications – all in one place.

[Job listings](#) [Mock interview](#) [Career planner](#) [Application tracker](#)

Answer in English only. Use a few sentences. Behavioral: use situation → action → result.

Question 1 of 5

Give an example of a goal you set and how you measured success.

Question 2 of 5

Describe a conflict on a team and how you resolved it.

Question 3 of 5

Tell me about a time you went above expectations.

Question 4 of 5

Tell me about a time you improved a process.

Question 5 of 5

Describe a situation where you had to persuade others.

[Finish & get feedback](#)

Recent sessions

[Clear history](#)

General Internship
Standard · Sun, Mar 22, 2026, 11:11 AM
[View session →](#)

SCORE
65%


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Fig. 20. Career page industry mock interview practice

35

Career
Explore internships, networking, and paths toward your first role after graduation.

Research **Industry**

Industry & careers
Browse live listings, plan your path, practice interviews, and track applications – all in one place.

Job listings Mock interview **Career planner** Application tracker

Your goal
Year focus: Internships, advanced projects, leadership, interview prep, and targeted applications.
Course ideas below use the **USC Columbia** catalog; always confirm prerequisites & sections in your official bulletin.

Target role
UX Designer

Class year
Junior

Skills (comma-separated)
Python, Git, Communication, Dance

The roadmap updates automatically a moment after you change role, year, or skills. Use refresh if something looks stale.

Refresh roadmap now

Progress 0%

Next best action – Close a gap – Figma basics
Open mock interview practice →

Milestones

Mark in progress	Close a gap – Figma basics Build a small library page with states, focus order, and contrast notes; pair with WCAG quick checks. Expected output: Documented components with accessibility notes. Deliverables: Component page All checklist	Immediate	Details & progress
Mark in progress	Network with design interns and alumni Ask about critique culture and portfolio expectations. Expected output: Documented networking with tailored hooks. Deliverables: Outreach log Story bank	Immediate	Details & progress
Mark in progress	Advanced HCI, visual design, or design ethics elective Pick a class that yields a portfolio-grade artifact. Expected output: Graded work tied to UX narrative. Deliverables: Course artifacts Resume bullets	This semester	Details & progress
Mark in progress	Extend that case into a small design system slice for the same product surface Document components, usage, and accessibility notes tied to the case study narrative. Expected output: Design system slice with documentation. Deliverables: System Figma Accessibility notes	This semester	Details & progress
Mark in progress	Lead design for a campus product or org Own roadmap for design deliverables; run critiques. Expected output: Leadership artifact with shipped work. Deliverables: Critique cadence Shipped assets	This semester	Details & progress
Mark in progress	Ship a full case study with metrics or outcomes Include constraints, research, iterations, and what you would measure if shipped. Expected output: Polished case with honest limits. Deliverables: Case study page Outcome or proxy metrics	This semester	Details & progress
Mark in progress	App critique + whiteboard + portfolio deep dive Practice live critiques of popular apps; rehearse deep dives on two projects. Expected output: Confidence across UX interview archetypes. Deliverables: Critique notes Deep dive outline	Next 6 months	Details & progress
Mark in progress	Apply to targeted UX internships with tailored portfolios Customize case order per company product area. Expected output: High-signal applications. Deliverables: Company map Portfolio variants	Next 6 months	Details & progress
Mark in progress	Return-offer or backup plan for senior year If interning, align on critique expectations; if not, plan fall projects. Expected output: Clear strategy for next 12 months. Deliverables: Manager sync notes Backup plan	Long term	Details & progress

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Fig. 21. Career page industry career planner roadmap

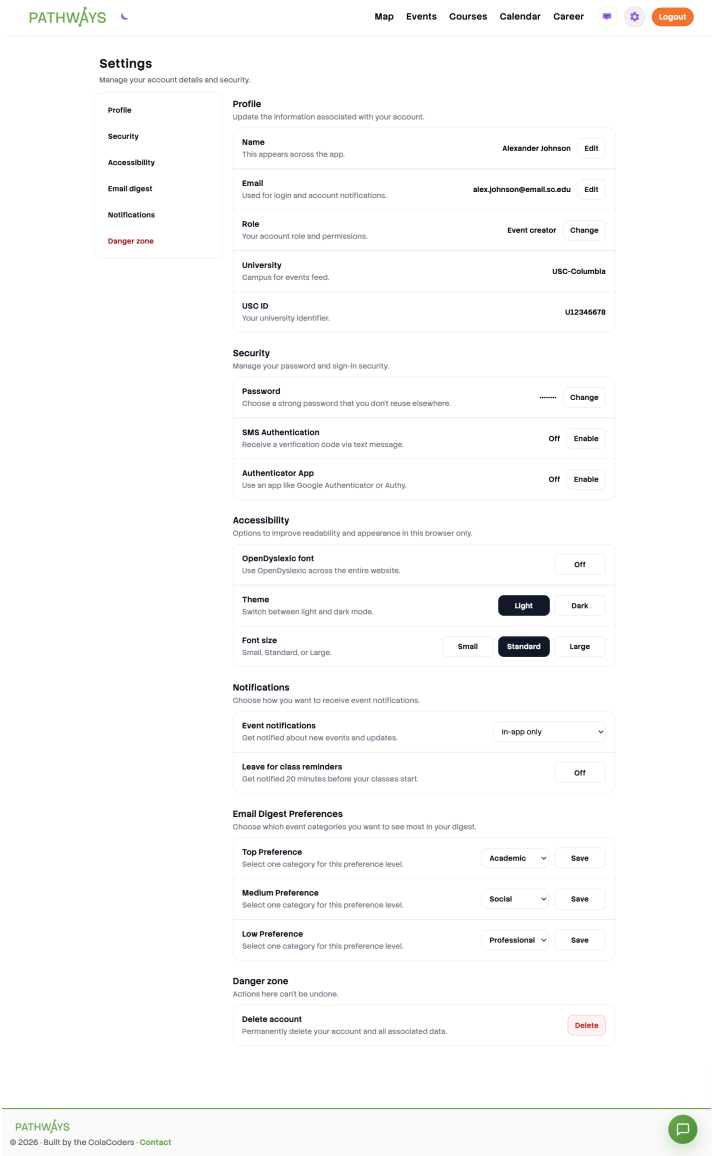


Fig. 22. Pathways settings page

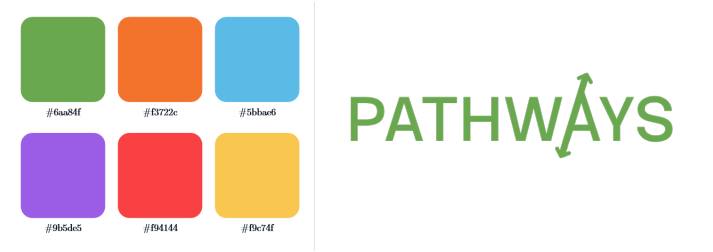


Fig. 23. Website color palette (left) and Pathways logo (right)